Day 23: Pick a way and prove it

Name _____ Date _____

I can pick a way to compare, and prove my answer.

Warm-up. Use $\frac{1}{2}$ to decide. Write >, = or <. $\frac{47}{100}$ ___ $\frac{7}{12}$ ____ because

_____

Example. Compare $\frac{2}{10}$ and $\frac{3}{4}$ three ways: with $\frac{1}{2}$, with the same parts, and with the same count.

With 1/2: 2/10 is _____ 1/2 and 3/4 is _____ 1/2, so 2/10 ____ 3/4.

Same parts: 2/10 = ____/20 and 3/4 = ____/20, so 2/10 ____ 3/4.

Same count: 2/10 = 6/____ and 3/4 = 6/____, so 2/10 ____ 3/4.

Cards. Say the sign and your way first. Prove your cards. On your partner's cards, write the sign. Is it a trap? Write "cannot tell" and why.

Card	**Sign, way and proof**
1. $\frac{5}{10}$ ___ $\frac{7}{12}$	
2. $\frac{2}{8}$ ___ $\frac{3}{12}$	
3. Jay read $\frac{1}{2}$ of his book. Kim read $\frac{1}{3}$ of her book. Who read more pages?	
4. $\frac{8}{12}$ ___ $\frac{4}{5}$	
5. $\frac{9}{8}$ ___ $\frac{8}{6}$	
6. $\frac{6}{8}$ ___ $\frac{8}{10}$	

More cards.

Card	Sign, way and proof
7. Mo drank $\frac{4}{8}$ of a carton of milk. Lu drank $\frac{1}{6}$ of another carton. Who drank more milk?	
8. $\frac{1}{4}$ — $\frac{2}{12}$	

Stories. Show or tell how you know.

1. Ivy's ribbon is $\frac{13}{10}$ meters long. Max's ribbon is $\frac{5}{3}$ meters long. Whose ribbon is longer? Write the comparison with >, = or <.

2. Noor ate $\frac{4}{6}$ of her sandwich. Sam ate $\frac{3}{5}$ of his sandwich. Noor says, "I ate more, because $\frac{4}{6} > \frac{3}{5}$." Can she be sure? Tell why.

Prove it. Which is greater, $\frac{5}{6}$ or $\frac{9}{10}$? Prove it two ways. One way may be a number line.